Matthew Kuney

State College, PA

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Education

The Pennsylvania State University **Expected May 2030**

Bachelor of Science in Mechanical Engineering *University Park, PA*

Relevant Coursework

- Calculus III
- Matrices

Experience

Progressive Pool Management Inc. **May 2024 – Aug 2025**

Head Lifeguard *Garnet Valley, PA*

- Managed independent 10-hour shifts, including facility opening and closing procedures.
- Oversaw pool safety, patron services, admissions, and daily facility operations.
- Maintained proper pool chemical levels and performed routine water testing.
- Responded to emergencies and enforced facility safety policies and procedures.

Mathnasium – The Math Learning Center **Sep 2025 – May 2026**

Math Instructor *Kennett Square, PA*

- Provided one-on-one and small-group mathematics instruction to students across multiple grade levels.
- Helped students develop problem-solving, analytical thinking, and mathematical reasoning skills.
- Communicated with students and staff to track progress and identify areas for improvement.

YMCA of Greater Brandywine **Jun 2026 – Aug 2026**

Camp Counselor *Kennett Square, PA*

- Supervised and supported campers in daily activities while maintaining a safe and positive environment.
- Planned and facilitated recreational, educational, and team-building activities.
- Managed groups of campers, maintained schedules, and addressed behavioral and interpersonal issues.

Projects

Autonomous Pick-and-Place Robot | *Arduino, CAD, Sheet Metal Fabrication, 3D Printing*

- Designed and built a drone-controller-operated pick-and-place robot from scratch, including a welded sheet-metal chassis, drive motors, and a servo-actuated arm.
- Modeled and 3D-printed a custom gripper mechanism to grasp and relocate objects.
- Integrated an Arduino microcontroller with a buck converter and battery system to power and control the robot's motors and electronics.

Fully Mechanical 3D-Printed Excavator Model | *Onshape CAD, 3D Printing*

- Designed and fabricated a fully mechanical, non-electronic excavator model in Onshape, driven entirely by springs, gears, nuts, and bolts.
- Engineered a 3-degree-of-freedom arm and bucket assembly, with the cab swiveling on a bevel gear drivetrain.

Bluetooth Tone-Control Speaker System | *3D Printing, Electronics*

- Rewired salvaged TV stereo speakers into a standalone Bluetooth speaker system with tone controls.
- Designed and 3D-printed a custom enclosure to house the speaker electronics.

Technical Skills

CAD & Design: Onshape
Fabrication: 3D Printing, Laser Cutting
Electronics: Arduino, Circuit Design
Other: Mechanical Design, Prototyping, Manufacturing

Leadership / Extracurricular

ASME (American Society of Mechanical Engineers) **2026 – Present**

Member *Penn State University*

- Helped organize and run Penn State ASME's Battle Bots competition.

Varsity Soccer **2023 – 2025**

Team Member *Garnet Valley High School*

- Competed at the varsity level from sophomore through senior year, balancing rigorous athletic commitments with academic coursework.